

Simulation of Paraffin Deposition in Reservoirs

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Summary. Certain reservoirs have a recurring wellbore damage problem caused by deposition of solid paraffin particles in the pore space. This paper describes the simulation of the reservoir processes of paraffin precipitation, convection, and deposition. The effect of reservoir heating treatments is included.

Introduction

Most paraffin solids are mixtures of hydrocarbons ranging from $C_{18}H_{38}$ to $C_{70}H_{172}$. Deposition of these thick, waxy substances in wellbores and production facilities can reduce operating efficiency. However, certain reservoirs also have paraffin deposition in the pore space, causing productivity damage. These reservoirs are the subject of this paper.

Fig. 1 shows the production profile of an actual well before and after heat treatment.¹ The production rate was initially very low because of paraffin damage in the formation. With stimulation (heating), dramatic increases in flow rates were observed. The well was treated periodically to maintain a high production rate because paraffin damage recurred between treatments.

Paraffin deposition in reservoirs consists of three mechanisms: (1) solubility/precipitation of paraffin, (2) transport and deposition of precipitated paraffin, and (3) reduction in permeability caused by paraffin deposition.

Reistle² conducted a comprehensive study on paraffin solubility. He reported that paraffin solubility decreases with decreases in temperature and in lighter constituents (gas) in the oil. Weingarten and Euchner³ found that paraffin solubility can be modeled with the ideal-solution theory of two components. They used a semi-empirical fit of the ideal-solution theory to match laboratory data that related cloud point (the equilibrium temperature and pressure at which solid paraffin crystals begin to form in the liquid phase) to the amount of dissolved gas during pressure depletion.

Solid paraffin particles flowing through the reservoir either clog pore throats or stick to pore walls (see Fig. 2). This is called deposition. Iwasaki⁴ introduced a simple and appropriate deposition model in 1937. Lambert⁵ and Civan and Knapp⁶ used empirical relationships to model the reduction in absolute permeability caused by deposition of solid particles. Their models represent reduced permeability as a function of reduced porosity.

The purpose of the research reported here was (1) to develop a numerical simulator to model the paraffin depositional and removal processes and (2) to improve our understanding of the effects of these processes on well productivity.

Simulator Development

Components and Phases. Fig. 3 illustrates the components used in the simulator, as well as the phases considered in the reservoir and on the surface. Three components (or pseudocomponents) were used: gas, oil, and paraffin. At surface conditions, the gas component comprises the produced gas. Stock-tank oil, however, is made up of the oil component plus the paraffin component.

At reservoir conditions, three phases are considered: vapor, liquid, and solid. The gas component comprises the vapor phases but also is dissolved in the liquid phase. The paraffin component comprises the solid phase but also is dissolved in the liquid phase. The oil component exists only in the liquid phases. Therefore, the liquid phase includes all three components.

Paraffin Solubility Model. Paraffin solubility/precipitation was modeled with the equation of ideal-solution theory for a binary mixture³:

$$\ln x_{ps} = \frac{-\Delta H_f}{R} \left(\frac{1}{T} - \frac{1}{T_m} \right) \quad (1)$$

This gives the paraffin solubility, x_{ps} (mole fraction), as a function of temperature, T . The parameters ΔH_f and T_m must be calibrated for a particular liquid/solid system. The paraffin component and gas plus oil components (in the liquid phase) are taken to be the binary mixture. In other words, the vapor phase is ignored. Thus, when gas leaves the liquid, the paraffin concentration increases.

Paraffin Deposition Model. Solid paraffin particles become entrapped in the pore throats as they flow through the reservoir, causing deposition. Without flow, no deposition occurs.

The following deposition equation⁴ states that the rate of loss of suspended solids along a flow path is equal to the concentration of suspended solids times a deposition coefficient:

$$dC/ds = -\lambda C \quad (2)$$

We can show that the expected (average) travel distance of a particle is $1/\lambda$. For example, if $\lambda = 0.5 \text{ ft}^{-1}$, then $E(s) = 2.0 \text{ ft}$. It was assumed that λ is constant in the simulator.

Permeability Reduction Model. The deposition of solid paraffin particles in the reservoir causes a permeability reduction. The permeability reduction ratio is related to the reduction in effective pore space in a form given by Civan⁶.

$$k/k_i = (1 - \sigma)^m \quad (3)$$

Eq. 3 uses a parameter, m , that must be calibrated to experimental results. Civan recommended a value of 3.0 for plugging by sand particles. As explained later, we found that $m = 8.0$ seemed to fit better for plugging by paraffin particles. Fig. 4 shows these two relationships.

Finite-Difference Equations. A standard finite-difference approach was used for fully implicit simulation of four equations in two dimensions. There are three mass-balance equations, one for each component, and an energy-balance equation.

The gas-component mass-balance equation is

$$\Delta a_{gv} \Delta \Phi_v + \Delta (R_{sg} a_{oL}) \Delta \Phi_L = \frac{V}{\Delta t} \delta (\phi \rho_v S_v + \phi R_{sg} \rho_{oL} S_L) + (\rho_g)_{sc} q_g \quad (4)$$

For the oil component,

$$\Delta a_{oL} \Delta \Phi_L = \frac{V}{\Delta t} \delta (\phi \rho_{oL} S_L) + (\rho_{oL})_{sc} q_o \quad (5)$$

The paraffin component is given by

$$\Delta (R_{ps} a_{oL} \Delta \Phi_L) + \Delta (R_{sp} a_{oL}) \Delta \Phi_L = \frac{V}{\Delta t} \delta (\phi \rho_s S_s + \phi R_{sp} \rho_{oL} S_L) + (\rho_{pL})_{sc} q_o \quad (6)$$

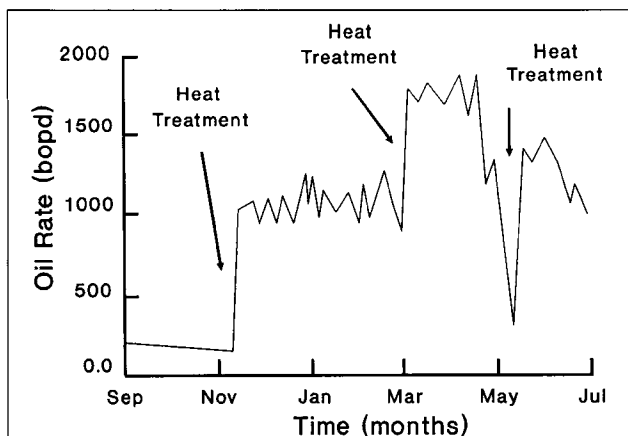


Fig. 1—Production performance of an actual well stimulated with heat treatments.

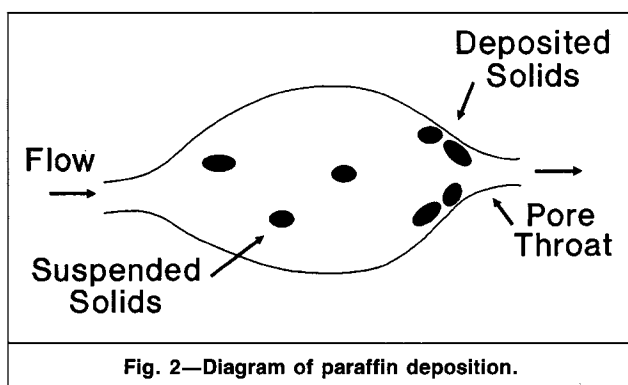


Fig. 2—Diagram of paraffin deposition.

The energy-balance equation is

$$\Delta a_h \Delta \Phi + \Delta a_c \Delta T = \frac{V}{\Delta t} \delta [\phi \rho_v S_v E_v + \phi \rho_L S_L E_L + \phi \rho_s S_s E_s + (1 - \phi) \rho_f E_f] + q_h + q_L - q_{EM} \quad (7)$$

The solid saturation is divided into suspended and deposited solids. The suspended solids flow at the same rate as the liquid phase; the deposited solids are immobile.

Gas Solubility and Natural Cooling. Gas solubility as a function of pressure and temperature, and the resulting expansion of oil, were calculated with Standing's correlations. Oil viscosity was calculated with the Beggs-Robinson⁷ correlation as a function of dissolved gas and dead-oil viscosity.

"Natural" cooling may take place near the wellbore. This cooling results from gas expansion and from gas evolution from solution. This temperature drop is calculated with Eq. 7. A method was developed to calculate the enthalpies of each phase as functions of pressure, temperature, and composition. This method was based on the Passut-Danner equation⁸ for calculating enthalpies resulting from temperature changes, and a Lee-Kesler-Pitzer equation⁸⁻¹⁰ was used to calculate enthalpy departures owing to changes in pressure.

Dissolved gas acts as a paraffin solvent. As pressure drops near the wellbore, gas leaves solution and causes paraffin precipitation. In addition, temperature decreases near the wellbore, causing further paraffin precipitation. The effects of temperature and pressure on paraffin solubility are computed with Eq. 1. Ref. 11 gives details.

Simulator Validation and Calibration

Validation of the fluid flow simulation was routine. The agreement with known cases was excellent if small grids and timesteps were used, as expected from finite-difference equations.

PSEUDO-COMPONENTS	PHASES			PHASES	
	Vapor	Liquid	Solid	Vapor	Liquid
Gas	X	X		X	
Oil		X			X
Paraffin		X	X		X
	RESERVOIR CONDITIONS			STOCK TANK CONDITIONS	

Fig. 3—Diagram of the components of each phase.

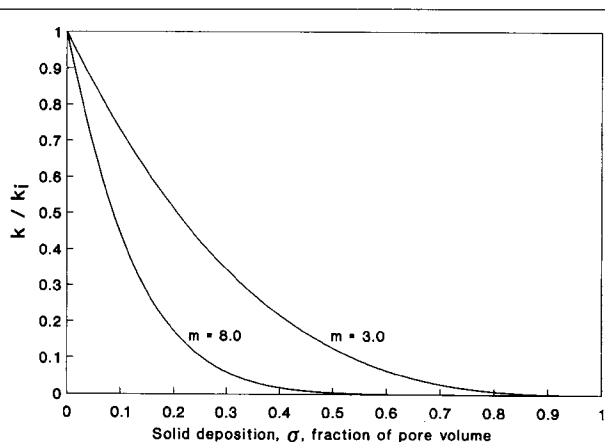


Fig. 4—Graphical representation of permeability reduction model, showing $m = 3.0$ for sand particles and $m = 8.0$ for paraffin particles.

Checking the solutions for solid particle transport and deposition (Eqs. 2 and 3) was more challenging. Analytical solutions were developed for three selected cases. The simulations of these three cases were compared with the analytical solutions. Comparisons showed that numerical dispersion of the suspended particle concentration, C , could be very significant near a front. This reminds us of our problems with numerical dispersion near a Buckley-Leverett front for immiscible displacement. However, the values for paraffin deposition behind the front, σ , were extremely accurate. The simulator was judged to be sufficiently accurate because paraffin damage depends on σ , not C . Ref. 1 gives details of these analytical solutions.

Calibration of Deposition and Permeability Reduction Models. The deposition coefficient, λ , of the deposition model and the constant, m , of the permeability reduction model were determined by matching the simulator with Sutton and Roberts'¹² laboratory results. The purpose of Sutton and Roberts' experiment was to determine the amount of permeability reduction that results from paraffin deposition in linear cores.

Values of λ and m were determined by matching simulated results with two of their experiments. Many data necessary to simulate the experiments were not available, so assumptions were made to simulate both experiments.

The first experiment involved flowing a Shannon sand crude oil through a Berea sandstone core. The temperature of the crude oil was set at 70°F, 30°F below its cloud point. The paraffin and asphaltene contents of the sample were 4.1 and 0.7 wt%, respectively. Based on these properties (indicative of the hydrocarbon component C₃₇), listed in Table 1, the solid saturation of the injected fluid was calculated to be 3.34 vol% at 70°F. Table 2 lists the grid

TABLE 1—FLUID PROPERTIES USED TO SIMULATE THE FIRST LABORATORY EXPERIMENT

Temperature, °F	70.0
Saturations, fraction	
Vapor	0.0
Liquid	0.9666
Solid	0.0334
Oil pseudocomponent	
Molecular weight, lbm/lbm-mol	104.11
Density, lbm/ft ³	44.69
Partial liquid density, lbm/ft ³	44.53
Paraffin pseudocomponent	
Molecular weight, lbm/lbm-mol	522.4
Density, lbm/ft ³	61.08
Partial liquid density, lbm/ft ³	0.21
Melting point, °R	628.2
Latent heat of fusion, kcal/g-mol	26.0

data, porosity and initial permeability of the Berea sandstone core, and initial flow rate used to simulate this experiment. Although the initial flow rate was assumed to be 0.38 ft³/D, the simulated results were not sensitive to the initial flow rate.

Fig. 5 compares the simulated and experimental results. A family of curves is shown for $\lambda=0.5, 0.6, 0.7$, and 0.8 . A value of $m=8.0$ was found to fit best and was used for all these curves.

The second experiment involved flowing a Muddy formation crude oil through a Berea sandstone core. The crude oil temperature was set at 70°F, about 25°F below its cloud point. The paraffin and asphaltene contents of the crude oil were 6.1 and 0.1 wt%, respectively. Based on these properties (indicative of the hydrocarbon component C₃₄), listed in **Table 3**, the solid saturation of the injected fluid was calculated to be 3.22 vol% at 70°F. **Table 4** gives other data used for simulating this experiment.

Fig. 6 compares the simulated and experimental results. A family of curves is shown for $\lambda=0.4, 0.5$, and 0.6 . Again, a value of $m=8.0$ was found to fit best and was used for all these curves.

A good match between the simulated and experimental results was not obtained for the total flow period of either experiment. However, an adequate match was achieved for the flow period between 2.5 and 5.0 PV of the second experiment. This match was obtained with values of $\lambda=0.5$ and $m=8.0$. It was not as important to match the initial flow period, between 0.0 and 2.5 PV, because most rate reduction occurred between 2.5 and 5.0 PV.

Although the matches of the laboratory data are not as good as we would have liked, values of $\lambda=0.5$ and $m=8.0$ were used throughout this study.

Results

Case 1 (Above Bubblepoint). This case was chosen to simulate paraffin damage and removal for a reservoir that initially has paraffin

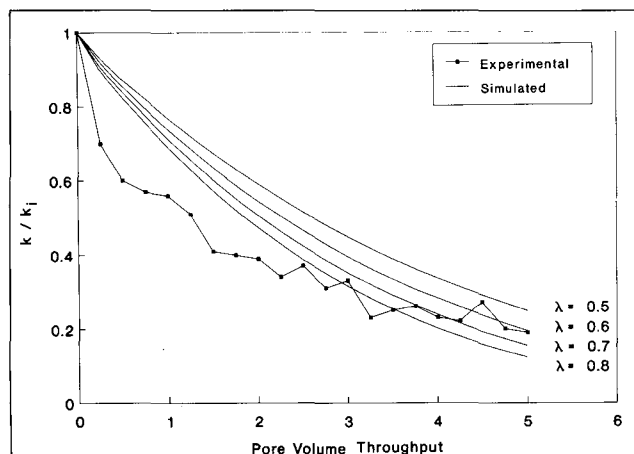


Fig. 5—Comparison of simulated and experimental results of Sutton and Roberts' first experiment.

TABLE 2—DATA USED TO SIMULATE THE FIRST LABORATORY EXPERIMENT

Δx , ft	0.0833
Δy , ft	0.1221
h , ft	0.1221
L , ft	1.0
Number of gridblocks	12
ϕ , fraction	0.25
k_i , md	405.0
q_i , ft ³ /D	0.38

particles but never has free gas. The reservoir data used to simulate this case were similar to those from the Schoonebeek field in The Netherlands. Although this field has been prolific, production rates in parts of the field seem to be low because of paraffin damage. The reservoir temperature, 100°F, is believed to be 4°F below the cloud point, resulting in a solid paraffin saturation at initial reservoir conditions.

The performance of a single well with constant outer boundary pressure was simulated with the data in **Table 5**. Paraffin content of the Schoonebeek oil was not available; therefore, four scenarios were simulated with oils having a paraffin content of $C_i=0.95, 3.50, 6.05$, and 8.60 vol% (equivalent to 5.0, 7.5, 10.0, and 12.5 wt%).

Fig. 7 shows the simulated oil rates. These results illustrate a rapid decrease in oil rates with time, which is typical of this reservoir. The stabilized rate, 62.5 STB/D, of the $C_i=3.50$ case matches the reduced rates of a Schoonebeek well, which was approximately 62.9 STB/D. This is about 50% of the undamaged rate. The deposition profiles in **Fig. 8** illustrate that most deposition occurs near the wellbore.

The heating process that was simulated to remove the deposited paraffin is known as electrical resistance heating (ERH).¹³ This noncontact thermal recovery method introduces heat into the formation and converts low-frequency electrical power to heat by the resistance of the formation to current flow. ERH was investigated for potential use in the Schoonebeek field.

Cases were simulated with various heating times at a constant power source of 20 kW. The total production period of each case was 730 days with heating times, t_h , of 1, 7, 30, 365, and 680 days. Each treatment started after 50 days of production, which corresponds to a flow rate of 64.5 STB/D.

Fig. 9 shows the simulated oil rates for each treatment compared with an untreated well. These results illustrate that a significant increase in oil rate occurs for each treatment. In fact, every case exhibits a permanent increase in oil rates because no new precipitation occurs.

The paraffin deposition profiles for the simulation runs in **Fig. 9** are shown in **Fig. 10** at a production time of 730 days. These

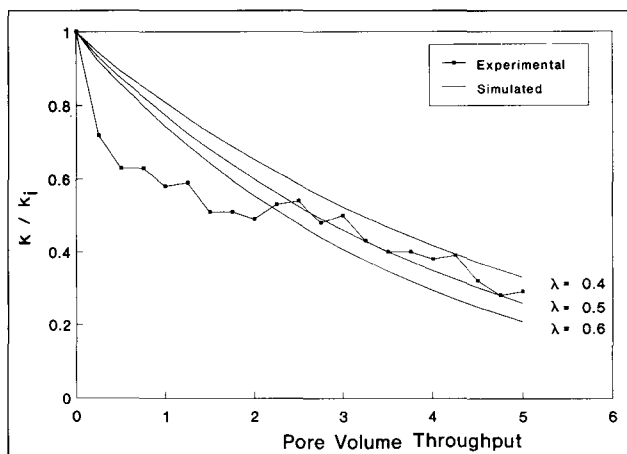


Fig. 6—Comparison of simulated and experimental results of Sutton and Roberts' second experiment.

TABLE 3—FLUID PROPERTIES USED TO SIMULATE THE SECOND LABORATORY EXPERIMENT

Temperature, °F	70.0
Saturations, fraction	
Vapor	0.0000
Liquid	0.9678
Solid	0.0322
Oil pseudocomponent	
Molecular weight, lbm/lbm-mol	122.51
Density, lbm/ft ³	46.71
Partial liquid density, lbm/ft ³	46.34
Paraffin pseudocomponent	
Molecular weight, lbm/lbm-mol	479.70
Density, lbm/ft ³	61.08
Partial liquid density, lbm/ft ³	0.49
Melting point, °R	621.00
Latent heat of fusion, kcal/g-mol	23.60

TABLE 4—DATA USED TO SIMULATE THE SECOND LABORATORY EXPERIMENT

Δx , ft	0.0833
Δy , ft	0.1221
h , ft	0.1221
L , ft	1.0
Number of gridblocks	12
ϕ , fraction	0.25
k_i , md	314.0
q_i , ft ³ /D	0.3

TABLE 5—DATA USED TO SIMULATE CASE 1

Reservoir properties	
Thickness, ft	45.9
Radial extent, ft	640.0
Permeability, md	7,500.0
Porosity, fracture	0.30
Temperature, °F	100.0
Initial pressure, psia	1,300.0
Initial stabilized oil rate, STB/D	118.0
Oil viscosity, cp	180.0
Gravity, °API	24.7
Oil pseudocomponent	
Molecular weight, lbm/lbm-mol	356.5
Density, lbm/ft ³	56.16
Paraffin pseudocomponent	
Molecular weight, lbm/lbm-mol	466.1
Density, lbm/ft ³	57.59
Melting point, °R	617.4
Latent heat of fusion, kcal/g-mol	22.8
Simulator data	
Number of gridblocks	14
Bottomhole flowing pressure, psia	1,233.0
Timestep, days	0.025
Permeability reduction constant, m	8.0
Deposition coefficient, λ , ft ⁻¹	0.5

results show that the deposited paraffin has been removed from the near-wellbore region (i.e., no additional precipitation has occurred after the heat treatments). This explains the permanent increase in production.

Case 2 (Below Bubblepoint). Paraffin deposition in the reservoir was then studied for a case below bubblepoint pressure. Free gas evolves continually, particularly near the wellbore. Initially, the reservoir is at its cloud point and $C_i = 0.0$. The initial paraffin content (in solution) was 8.0 wt%. The performance of a single well with constant outer boundary pressure was simulated with the data in Table 6. This reservoir is hypothetical, but we tried to use realistic data.

Fig. 11 illustrates the temperature profiles in the reservoir during natural depletion. The temperature around the wellbore dropped

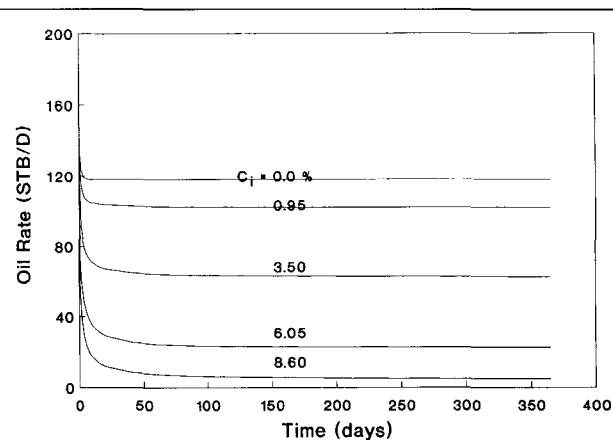


Fig. 7—Simulated oil rates for the Case 1 well showing the effect of initial solid concentration.

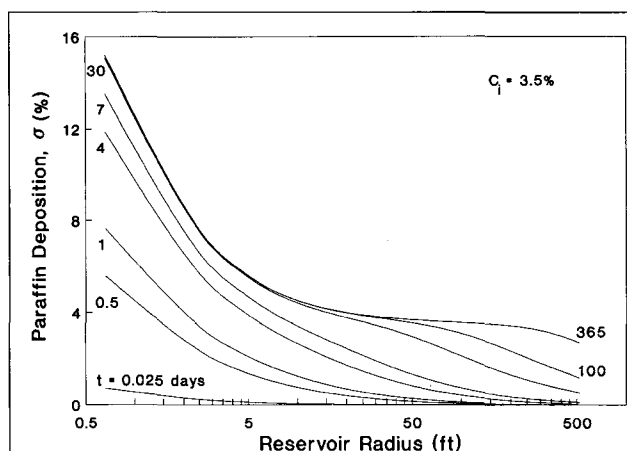


Fig. 8—Paraffin deposition profiles after production begins.

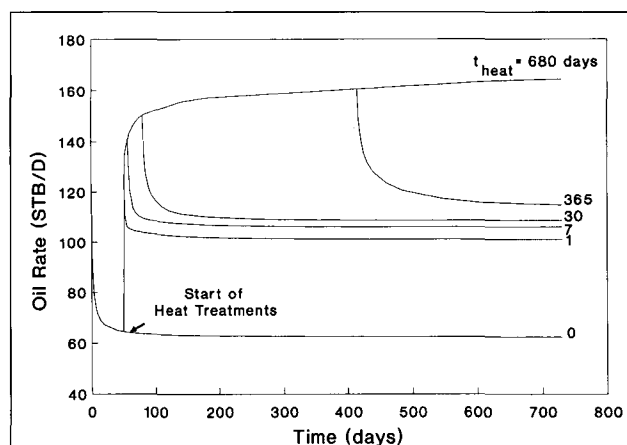


Fig. 9—Simulated oil rates for the Case 1 well treated with heat.

from an initial value of 100 to 87°F. This cooling results from gas expansion and gas evolution from solution. The reduced temperature causes paraffin precipitation.

Paraffin precipitates from gas evolution and from cooling. Simulation runs were made to separate these two factors. Fig. 12 plots oil rate as a function of time for three different reservoir scenarios: (1) no paraffin, (2) paraffin precipitation owing to gas evolution only at constant temperature, and (3) paraffin precipitation caused

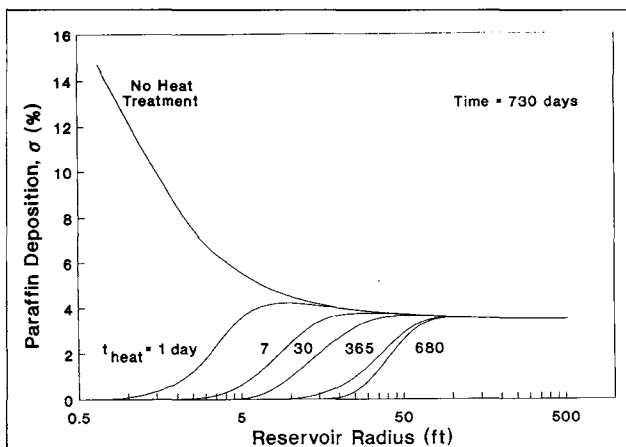


Fig. 10—Paraffin deposition profiles for a production time of 730 days.

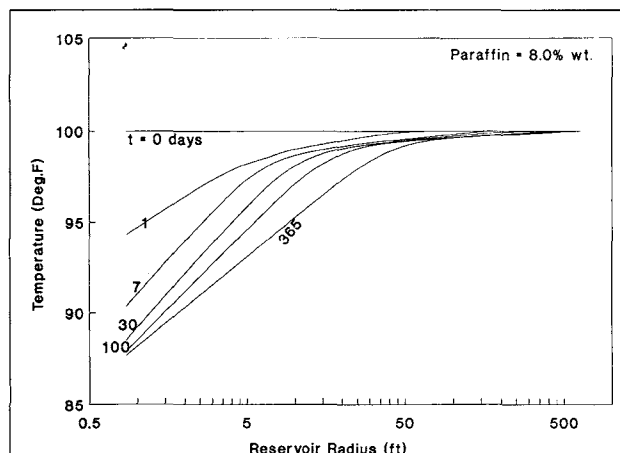


Fig. 11—Temperature profile for the Case 2 well showing temperature drop from gas expansion and evolution near wellbore.

TABLE 6—DATA USED TO SIMULATE CASE 2

Reservoir properties	
Thickness, ft	50.0
Radial extent, ft	640.0
Permeability, md	50.0
Porosity, fracture	0.3
Temperature, °F	100.0
Initial pressure, psia	2,000
Rock compressibility, psi^{-1}	20×10^{-6}
Gas pseudocomponent	
Gas gravity	0.75
Initial vapor saturation, fracture	0.06
Initial GOR, scf/stb	406.00
Oil pseudocomponent	
Molecular weight, lbm/lbm-mol	356.50
Density, lbm/ft ³	56.16
Compressibility, psi^{-1}	6×10^{-6}
Initial live oil viscosity, cp	64.0
Paraffin pseudocomponent	
Molecular weight, lbm/lbm-mol	466.10
Density, lbm/ft ³	58.61
Melting point, °F	186.20
Latent heat of fusion, kcal/g-mol	30.23
Simulator data	
Number of gridblocks	14
Bottomhole flowing pressure, psia	1,000.0
Permeability reduction constant, m	8.0
Deposition coefficient, λ , ft^{-1}	0.5

by both gas evolution and cooling. For the reservoir under consideration, temperature drop was a more important contributor to the paraffin problem than gas evolution.

Fig. 13 illustrates paraffin removal using cyclic heating with ERH. It plots oil rate as a function of time with $t_h = 7$ days at a time interval of 100 days. The first heating cycle is at 50 days. Three cycles were used.

Discussion

The simulation results show that solid paraffin particles do not migrate far before deposition. Consequently, solid particles that form farther away than 10 to 20 ft from the wellbore do not add much to the skin damage. With our value of $\lambda = 0.5$, one-half the suspended particles become deposited (stuck) when they travel only 1.4 ft. Only 1% of the particles travel 9.2 ft from their origin. The average travel distance is 2.0 ft. However, the converging nature of radial flow creates a significant paraffin buildup near the wellbore and reduction in well productivity.

The Case 1 simulation seemed to behave like the Schoonebeek well used to test the ERH process. With heat treatments, the oil rate increased to about the level reached in the simulation. The simu-

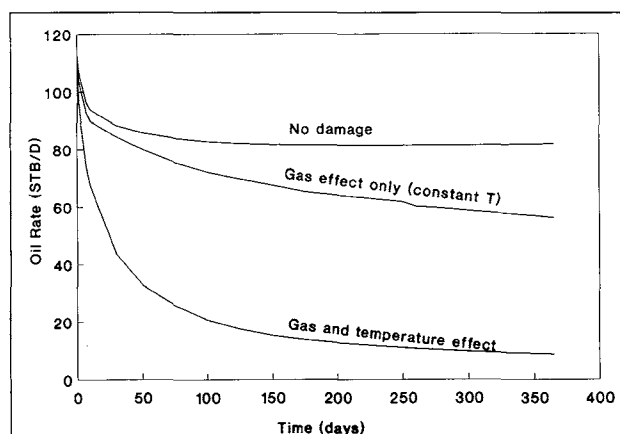


Fig. 12—Producing rate for the Case 2 well showing simulation runs with no precipitation (no damage) and constant temperature (gas effect only).

lation shows that paraffin removal would be permanent as long as the reservoir remained above the bubblepoint. The additional effect of reduced viscosity would eventually vanish after the treatment, of course, as the reservoir cools.

Installing an electrode for ERH in the wellbore can be costly. However, once the electrode is installed, the cost of ERH is relatively low. For example, the electric power of the 20-kW power source used in this simulation costs only about \$19/D, assuming a power cost rate of \$0.04/kW-hr. Once a well is completed with the electrode, it should be very economical to use ERH intermittently, whenever a reduction in production rate is noticed.

Case 2 shows a more probable well. No solid paraffin exists initially, but paraffin precipitation and deposition occur near the wellbore as gas evolves. This is a continuous process, unlike the case above the bubblepoint. Every barrel of oil carries dissolved paraffin toward the wellbore, waiting to precipitate. Consequently, it is necessary to remove paraffin damage periodically or continuously to restore oil rates.

Conclusions

1. The approach taken to model the deposition and removal processes of paraffin in the reservoir is adequate, given the limited laboratory data available (i.e., the simulator seems to give plausible results).
2. Solid paraffin particles do not travel far. Skin damage is caused by solid particles that form near the wellbore.

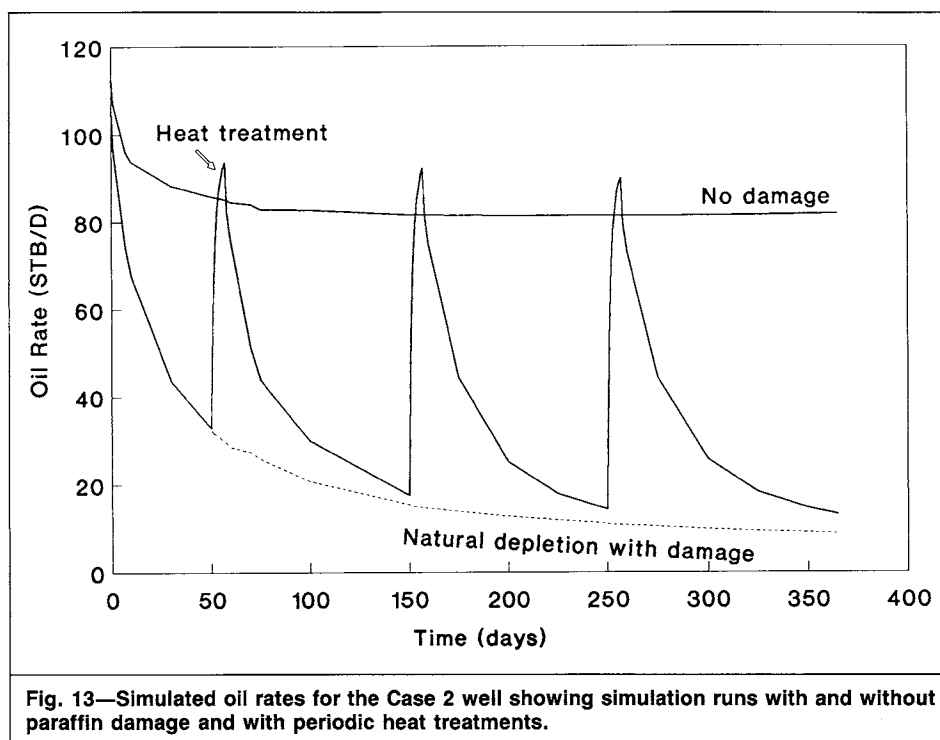


Fig. 13—Simulated oil rates for the Case 2 well showing simulation runs with and without paraffin damage and with periodic heat treatments.

3. ERH very rapidly removes most skin damage from solid paraffin buildup. Longer heating periods remove paraffin damage farther from the wellbore.

4. Gas expanding and evolving from solution near the wellbore cause significant cooling, a major factor in paraffin damage.

5. If gas evolves from solution near the wellbore, paraffin damage can be continuous. In this case, removal must be either continuous or periodic to restore oil rates.

Acknowledgments

We thank State Lands Energy Resource Optimization (SLERO) and the Dept. of Petroleum Engineering at Texas A&M U. for the support provided during the realization of this work. Thanks also go to Olmedo Jaimes for his work on the analytical solutions.

Nomenclature

A = cross-sectional area, L^2 , ft^2
 a_c = coefficient for conductive heat flow, mL^2/Tt^3 , $Btu/^\circ F-D$
 a_{gv} = coefficient for gas component flow in vapor phase, Lt , $lbm/psi-D$
 a_{ol} = coefficient for oil component flow in liquid phase, Lt , $lbm/psi-D$
 C = volume of suspended solids per volume of liquid plus suspended solids, dimensionless, ft^3/ft^3
 C_i = initial volume of suspended solids per volume of liquid plus suspended solids, dimensionless, ft^3/ft^3
 E_f = specific internal energy of the formation, L^2/t^2 , Btu/lbm
 E_L = specific internal energy of liquid phase, L^2/t^2 , Btu/lbm
 E_s = specific internal energy of solid phase, L^2/t^2 , Btu/lbm
 E_v = specific internal energy of vapor phase, L^2/t^2 , Btu/lbm
 h = height, L , ft
 ΔH_f = latent heat of paraffin fusion, L^2/t^2 , $cal/g-mol$
 k = absolute permeability, L^2 , md
 k_i = initial absolute permeability, L^2 , md
 L = length, L , ft

m = empirical constant for Eq. 3
 q_{EM} = power input from electromagnetic energy source, mL^2/t^3 , Btu/D
 q_g = flow rate of gas component, L^3/t , scf/D
 q_h = enthalpy rate of produced fluid, mL^2/t^3 , Btu/D
 q_L = rate of heat loss to overburden and underburden, mL^2/t^3 , Btu/D
 q_o = flow rate of oil plus paraffin components, L^3/t , scf/D
 r = radius, L , ft
 R = ideal constant, $L^2/t^2 T$, $1.987 \text{ cal/g-mol}^\circ K$
 R_{ps} = mass ratio of paraffin component suspended in the liquid phase to oil component in liquid phase, dimensionless
 R_{sg} = mass ratio of gas component dissolved in the liquid phase to oil component in liquid phase, dimensionless
 R_{sp} = mass ratio of paraffin component dissolved in the liquid phase to oil component in liquid phase, dimensionless
 s = travel distance along flow path, L , ft
 S_L = saturation of liquid phase, dimensionless, fraction
 S_s = saturation of solid phase, dimensionless, fraction
 S_v = saturation of vapor phase, dimensionless, fraction
 t = time, t , $days$
 t_h = heating time, t , $days$
 T = reservoir temperature, T , $^\circ R$
 T_m = melting-point temperature of paraffin, T , $^\circ R$
 v = linear velocity of liquid plus suspended solids, L/t , ft/D
 V = bulk volume, L^3 , $res \text{ ft}^3$
 x_{ps} = maximum amount of paraffin soluble in liquid phase, dimensionless, mole fraction
 λ = fraction of suspended solids removed per unit length, L^{-1} , ft^{-1}
 ρ_g = density of gas component, m/L^3 , $lbm/res \text{ ft}^3$
 ρ_L = density of liquid phase, m/L^3 , $lbm/res \text{ ft}^3$
 ρ_{oL} = density of oil component in the liquid phase, m/L^3 , $lbm/res \text{ ft}^3$
 ρ_{pL} = density of paraffin component in the liquid phase, m/L^3 , $lbm/res \text{ ft}^3$

ρ_s = density of solid phase, m/L³, lbm/res ft³
 ρ_v = density of vapor phase, m/L³, lbm/res ft³
 σ = volume of deposited solids per PV dimensionless, ft³/ft³
 ϕ = porosity, dimensionless, fraction
 Φ_L = potential gradient of liquid phase, m/Lt², psi
 Φ_v = potential gradient of vapor phase, m/Lt², psi

Subscripts

i = initial
sc = standard conditions

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SI Metric Conversion Factors

$^{\circ}\text{API} \quad 141.5/(131.5 + ^{\circ}\text{API}) = \text{g/cm}^3$
 $\text{bbl} \times 1.589\,873 \quad \text{E}-01 = \text{m}^3$
 $\text{cal} \times 4.186\,8^* \quad \text{E}+00 = \text{J}$
 $\text{cp} \times 1.0^* \quad \text{E}+00 = \text{mPa}\cdot\text{s}$

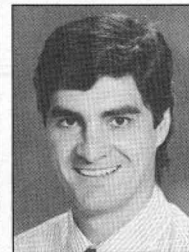
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ft × 3.048	E-01 = m
ft ³ × 2.831 685	E-02 = m ³
lbm × 4.535 924	E-01 = kg
md × 9.869 233	E-04 = μm ²
psi × 6.894 757	E+00 = kPa
°R °R/1.8	= K

*Conversion factor is exact.

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Original SPE manuscript received for review March 30, 1992. Revised manuscript received Nov. 10, 1993. Paper accepted for publication Nov. 12, 1993. Paper (SPE 24069) first presented at the 1992 SPE Western Regional Meeting held in Bakersfield, March 30-April 1.